

Image-Identifiable Genomic Subspaces

A Linear-Gaussian Model of Radiogenomic Recoverability

Joseph Rich Lior Pachter

Setup and notation

$g \in \mathbb{R}^p, x \in \mathbb{R}^d$	genomic / imaging feature vectors
$G \in \mathbb{R}^{n \times p}, X \in \mathbb{R}^{n \times d}$	stacked data (n patients)
$A \in \mathbb{R}^{d \times p}$	genomics-to-imaging transfer map
σ^2	isotropic imaging noise variance
$\Sigma_G, \Sigma_X, \Sigma_{GX}$	genomic / imaging / cross covariance
$\Sigma_{G X}$	posterior covariance (recoverability floor)
$B = \Sigma_{GX} \Sigma_X^{-1}$	Bayes-optimal map $x \mapsto \mathbb{E}[g \mid x]$
$M = \sigma^{-1} A \Sigma_G^{-1/2}$	whitened transfer operator; d_i its sing. values
$\rho_i, R_i = \rho_i^2$	i -th canonical correlation / recoverability
a_i, b_i	i -th canonical genomic / imaging direction
$I(G; X)$	mutual information between modalities
k	dimension of image-identifiable genomic subspace

The linear–Gaussian generative model

$$\begin{aligned} G &\sim \mathcal{N}(0, \Sigma_G), \\ X \mid G &\sim \mathcal{N}(AG, \sigma^2 I_d), \quad A \in \mathbb{R}^{d \times p}. \end{aligned}$$

Equivalently,

$$X = AG + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I_d), \quad \varepsilon \perp G.$$

Simplest model with a non-trivial answer: *every* downstream quantity is closed-form, so it acts as an exact upper bound on what any estimator can recover.

Joint Gaussian law

$$\begin{pmatrix} G \\ X \end{pmatrix} \sim \mathcal{N}\left(0, \begin{pmatrix} \Sigma_G & \Sigma_G A^\top \\ A \Sigma_G & A \Sigma_G A^\top + \sigma^2 I_d \end{pmatrix}\right).$$

In particular,

$$\begin{aligned} X &\sim \mathcal{N}(0, \Sigma_X), & \Sigma_X &= A \Sigma_G A^\top + \sigma^2 I_d, \\ \Sigma_{GX} &= \text{Cov}(G, X) = \Sigma_G A^\top. \end{aligned}$$

Imaging covariance splits additively into a genomics-driven part $A \Sigma_G A^\top$ and an irreducible part $\sigma^2 I_d$.

Joint law — proof sketch

(G, X) is an affine map of independent Gaussians (G, ε) , hence jointly Gaussian; both are centered, so the mean is 0.

$$\text{Cov}(G, G) = \Sigma_G,$$

$$\text{Cov}(G, X) = \text{Cov}(G, AG + \varepsilon) = \text{Cov}(G, AG) = \Sigma_G A^\top,$$

$$\text{Cov}(X, X) = \text{Cov}(AG) + \text{Cov}(\varepsilon) = A\Sigma_G A^\top + \sigma^2 I_d.$$

Cross terms vanish by $G \perp \varepsilon$. Collecting the blocks gives the joint covariance.



Bayes-optimal recovery: the posterior

$G \mid X \sim \mathcal{N}(\hat{g}(X), \Sigma_{G|X})$ with

$$\hat{g}(X) = \mathbb{E}[G \mid X] = \Sigma_G A^\top (A \Sigma_G A^\top + \sigma^2 I_d)^{-1} X = BX,$$

$$\Sigma_{G|X} = \Sigma_G - \Sigma_G A^\top (A \Sigma_G A^\top + \sigma^2 I_d)^{-1} A \Sigma_G.$$

Precision form (Woodbury):

$$\Sigma_{G|X}^{-1} = \Sigma_G^{-1} + \sigma^{-2} A^\top A, \quad B = \sigma^{-2} \Sigma_{G|X} A^\top.$$

$\Sigma_{G|X}$ is a **hard floor** on residual genomic uncertainty: no algorithm can do better. Imaging adds precision $\sigma^{-2} A^\top A$ to the prior.

Mutual information and the SNR decomposition

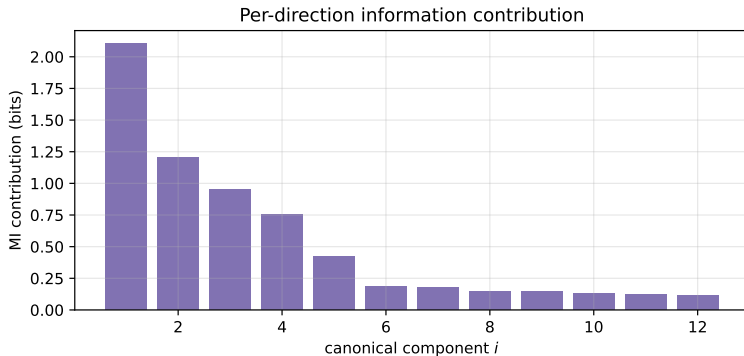
$$I(G; X) = \frac{1}{2} \log \frac{\det \Sigma_G}{\det \Sigma_{G|X}} = \frac{1}{2} \log \det(I_d + \sigma^{-2} A \Sigma_G A^\top).$$

With $\lambda_1 \geq \dots \geq \lambda_r > 0$ the nonzero eigenvalues of $A \Sigma_G A^\top$:

$$I(G; X) = \frac{1}{2} \sum_{i=1}^r \log \left(1 + \frac{\lambda_i}{\sigma^2} \right).$$

Each image-visible genomic direction contributes $\frac{1}{2} \log(1 + \text{SNR}_i)$. Directions with small signal relative to noise contribute nothing.

A handful of directions carry almost all the information



Per-direction $\frac{1}{2} \log(1 + d_i^2)$ on a synthetic instance ($n=800$, $p=80$, $d=40$); cumulative MI saturates at ≈ 6.45 bits after the first few components.

Recoverability = squared canonical correlation

Whiten $U = \Sigma_G^{-1/2} G$, transfer operator $M = \sigma^{-1} A \Sigma_G^{1/2} = \sum_i d_i u_i v_i^\top$. Canonical direction $a_i = \Sigma_G^{-1/2} v_i$, score $S_i = a_i^\top G$:

$$\text{Var}(S_i \mid X) = \frac{1}{1 + d_i^2}, \quad \rho_i = \frac{d_i}{\sqrt{1 + d_i^2}}.$$

$$R(a_i) = 1 - \text{Var}(S_i \mid X) = \frac{d_i^2}{1 + d_i^2} = \rho_i^2.$$

The whole analysis reduces to CCA between G and X : only the moments $\Sigma_G, \Sigma_X, \Sigma_{GX}$ enter — A and σ^2 never appear individually.

Recoverability of a named genomic direction

For a fixed score $Y = v^\top G$ (a gene, pathway, polygenic score):

$$R(v) = 1 - \frac{v^\top \Sigma_{G|X} v}{v^\top \Sigma_G v} \in [0, 1].$$

Equals the population R^2 of the Bayes-optimal predictor of Y from X . The core matrix has a moment-only form:

$$\Sigma_G - \Sigma_{G|X} = \Sigma_G A^\top \Sigma_X^{-1} A \Sigma_G = \Sigma_{GX} \Sigma_X^{-1} \Sigma_{XG}.$$

$R(v) = 0$: imaging carries no information about Y . $R(v) = 1$: imaging determines it exactly.

The recoverability eigenproblem = CCA

$$\max_{v \neq 0} R(v) = \max_{v \neq 0} \frac{v^\top (\Sigma_{GX} \Sigma_X^{-1} \Sigma_{XG}) v}{v^\top \Sigma_G v}$$

has stationary points solving the generalized eigenproblem

$$\Sigma_{GX} \Sigma_X^{-1} \Sigma_{XG} v = \mu \Sigma_G v, \quad \mu_i = \rho_i^2, \quad v = a_i.$$

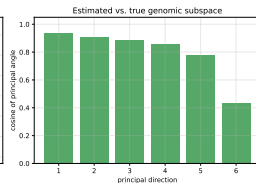
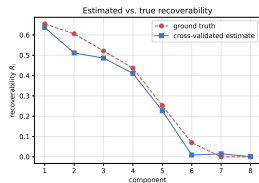
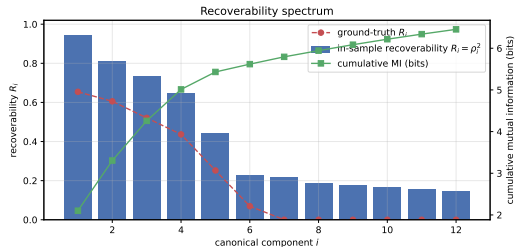
Solutions are exactly the canonical correlation directions, and $\max_v R(v) = \rho_1^2$. Deliverable: a *recoverability spectrum* $\rho_1^2 \geq \rho_2^2 \geq \dots$ with genomic directions $\{a_i\}$.

Rank limit and identifiability

$$\text{rank}(\Sigma_G - \Sigma_{G|X}) = \text{rank}(A\Sigma_G A^\top) \leq \min(d, p, \text{rank } A, \text{rank } \Sigma_G).$$

- ▶ At most $\min(d, p, \text{rank } A, \text{rank } \Sigma_G)$ canonical correlations are nonzero.
- ▶ Even with $p = 20,000$ genes, imaging exposes only a projection whose dimension is capped by the channel — not the gene panel size.
- ▶ Imaging features $x = \phi(\text{image})$ are themselves lossy: post-processing can only shrink $\text{rank}(\Sigma_{GX})$ and lower each ρ_i (data-processing inequality).

The estimator recovers the planted canonical structure



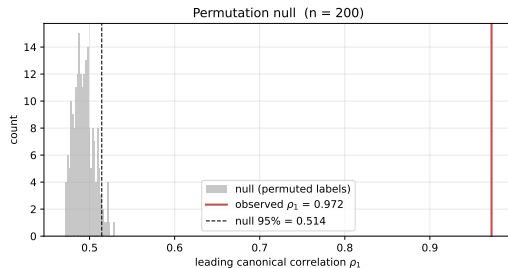
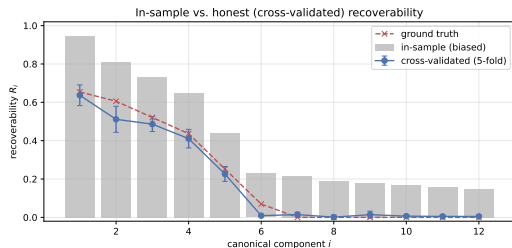
Synthetic instance with known truth: leading $\hat{\rho}_i^2$ matches the planted values (left); estimated quantities track the closed-form predictions (right).

Finite-sample regimes ($\kappa_G = p^*/n$, $\kappa_X = d^*/n$)

1. **Classical** ($\kappa_G + \kappa_X \ll 1$): sample CCA consistent, $\hat{\rho}_i \rightarrow \rho_i$ at rate $n^{-1/2}$.
2. **High-dimensional** ($\kappa_G + \kappa_X \lesssim 1$): null squared correlations follow a Wachter law; leading $\hat{\rho}_i$ inflated. Use regularized CCA.
3. **Saturating** ($p^* + d^* \geq n$): $\hat{\rho}_1 = \dots = 1$ regardless of association; spectrum not estimable.

Working rule: reduce *both* modalities so $p^* + d^* \leq n/\tau$, $\tau \in [5, 10]$. Report cross-validated $\hat{R}_i^{\text{cv}}$ and a scale-matched permutation null.

In-sample recoverability is inflated; CV + permutation are honest



In-sample $\hat{R}_i$ overshoots; 5-fold $\hat{R}_i^{\text{cv}}$ tracks the planted truth (left). Permutation null recovers the planted rank, $p < 0.01$ for the top three (right).

Estimation: regularized CCA

Shrinkage covariances (Ledoit–Wolf) $\hat{\Sigma}_G, \hat{\Sigma}_X$ and $\hat{\Sigma}_{GX} = n^{-1}G^\top X$. Then

$$\hat{M} = \hat{\Sigma}_G^{-1/2} \hat{\Sigma}_{GX} \hat{\Sigma}_X^{-1/2} = \hat{U} \hat{D} \hat{V}^\top,$$

$$\hat{\rho}_i = \hat{d}_i, \quad \hat{R}_i = \hat{\rho}_i^2, \quad \hat{a}_i = \hat{\Sigma}_G^{-1/2} \hat{u}_i, \quad \hat{b}_i = \hat{\Sigma}_X^{-1/2} \hat{v}_i.$$

Gaussian-copula marginal transform $\tilde{M}_{ij} = \Phi^{-1}(\text{rank}_i/(n+1))$ applied to both sides — radiomic marginals have excess kurtosis in the tens.

Bernoulli extension for mutations (Gaussian copula)

Latent $Z \sim \mathcal{N}(0, \Sigma_Z)$, $\text{diag}(\Sigma_Z) = I_p$; thresholding $G_j = 1\{Z_j > \tau_j\}$, $\tau_j = \Phi^{-1}(1 - p_j)$. Imaging couples to the latent: $X \mid Z \sim \mathcal{N}(BZ, \sigma^2 I_d)$.

Data-processing bound, tight up to the binarization gap:

$$I(G; X) \leq I(Z; X) = -\frac{1}{2} \sum_{i=1}^k \log(1 - \rho_i^2).$$

Per-gene AUC ceiling: $\text{AUC}_j \leq \Phi(\sqrt{R_j^Z/(1 - R_j^Z)}/\sqrt{2})$. All CCA / MI machinery transfers with $\Sigma_G \rightarrow \Sigma_Z$.

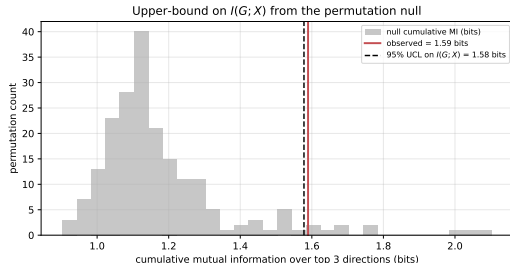
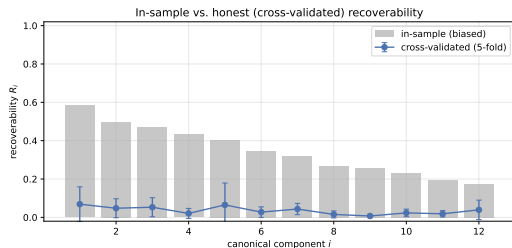
Application: TCGA-KIRC ($n = 190$)

Organ radiomics ($d_{\text{native}} = 107$), 2000 HVGs, copula marginals, $p^* = d^* = 38$.

- ▶ In-sample $\hat{\rho}_1 \approx 0.76$ ($\hat{R}_1 \approx 0.58$), 3.9 bits.
- ▶ Cross-validated $\hat{R}_1^{\text{cv}} \approx 0.07\text{--}0.1$; variance share $0.27 \rightarrow 0.011$.
- ▶ Permutation $p < 0.05$ for leading-3 in 12/12 preprocessing variants.
- ▶ $I(G; X)|_{\text{top-3}} \leq 1.58$ bits (one-sided 95% UCL).
- ▶ Effective rank $\hat{k}_{\text{eff}} = 1$ (median).

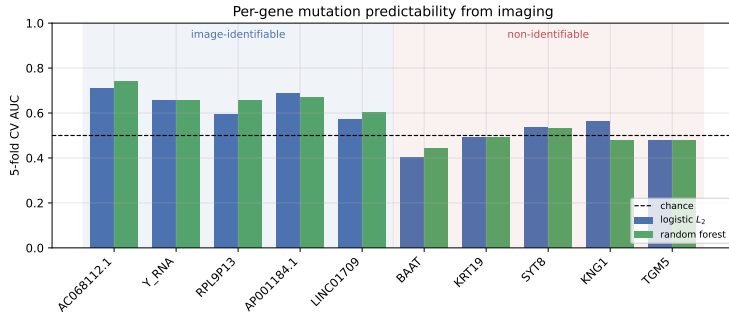
One genomic direction is robustly image-identifiable; recoverability ≈ 0.1 .

TCGA-KIRC: one weak but real image-identifiable direction



Cross-validated $\hat{R}_i$ is a small positive residual (left); observed cumulative MI sits at the upper edge of the null, $I(G; X) \leq 1.58$ bits (right).

KIRC: recoverability score predicts which genes imaging can read



High-recoverability genes clear the chance line ($AUC \approx 0.65$); low-recoverability genes sit at 0.5. Random forest does not exceed the linear ceiling.

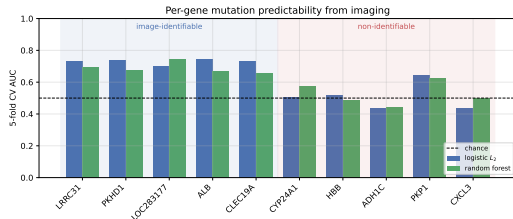
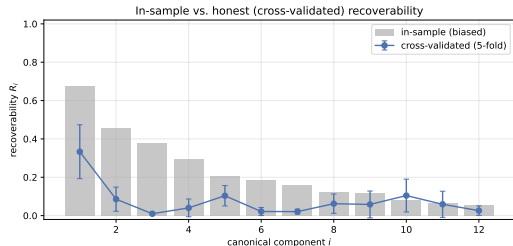
Application: NSCLC ($n = 90$) — the counterpoint

Same pipeline, lung organ radiomics, $p^* = d^* = 18$.

- ▶ In-sample $\hat{p}_1 = 0.82$, 2.6 bits; null well-separated ($p = 0.005$).
- ▶ Held-out canonical $R^2 = 0.40$; $\hat{R}_1^{cv} = 0.33$.
- ▶ Per-gene AUC: identifiable 0.73 / 0.69 vs non-identifiable 0.51 / 0.53.
- ▶ Survives de-confounding (scanner / kernel): $\hat{R}_1^{cv} \approx 0.24$ –0.30.

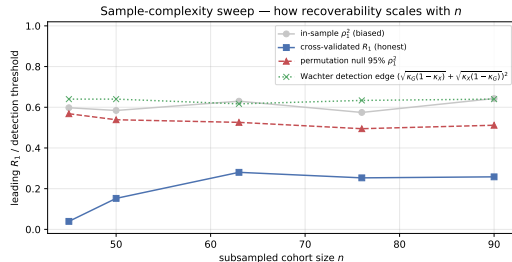
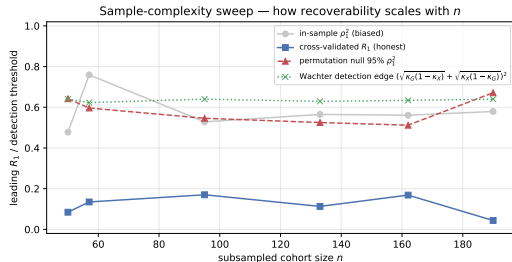
Two cohorts, same model: borderline single direction (≈ 0.1) in RCC, solid single direction (≈ 0.33) in NSCLC — same low rank, different strength.

NSCLC: a solid single direction, an order of magnitude above KIRC



Leading $\hat{R}_1^{\text{cv}} \approx 0.33$ (left); per-gene AUC 0.73/0.69 for identifiable vs 0.51/0.53 for non-identifiable genes (right).

Honest signal lives below the in-sample bulk at all cohort sizes



Sample-complexity sweeps, KIRC (left) and NSCLC (right): $\hat{R}_1^{\text{cv}}$ is visible only through CV/permutation, never the raw in-sample spectrum.

Takeaway

- ▶ Radiogenomics should target **image-identifiable low-dimensional genomic subspaces**, not individual genes.
- ▶ Recoverability of canonical direction i is exactly ρ_i^2 — computable by regularized CCA from $\Sigma_G, \Sigma_X, \Sigma_{GX}$ alone.
- ▶ The recoverable dimension is a property of the *biology–imaging channel*, bounded by $\text{rank}(A)$, not by panel size.
- ▶ Honest estimands are cross-validated and permutation-calibrated; in-sample spectra are badly inflated when $p, d \sim n$.